AAYUSHI KAPOOR

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EXPERIENCE

Software Developer Intern - Hewlett Packard Enterprise (HPE) *Jun 2026 - Aug 2026 | Bengaluru, India*

- Collaborated in a team of 3 to build Agentic Workflow Studio, a drag-and-drop platform with 12+ customizable node types and DAG-based execution powered by LangGraph, enabling reusable AI workflows and reducing workflow setup effort by ~60%.
- Developed an Employee Registration & Approval Platform using Python and SQL (PostgreSQL), leveraging relational databases to automate the provisioning of 500+ accounts and reduce onboarding time by over 90%.
- Engineered 'Port Observatory,' a data engineering solution using FastAPI and TimescaleDB, creating robust data pipelines to perform real-time data analysis on metrics from 100+ network ports, reducing manual monitoring effort by ~70%.

Research Intern - Defense Research and Development Organization (DRDO) *May 2025 - Jun 2025 | Delhi, India*

- Built a secure offline Flask + SQLite dashboard to manage and search 1,000+ research files in a restricted environment, reducing file retrieval time by ~50%. Designed a lightweight offline-first architecture with secure local storage and role-based access, deployed with zero cloud dependencies.
- Authored a comprehensive review paper on AI/ML applications in Brain-Computer Interfaces (BCI) in Defense, synthesizing existing research and emerging techniques; the work was reviewed and appreciated by DRDO scientists.

EDUCATION

Delhi Skills and Entrepreneurship University (DSEU) - Delhi, India *Expected 2027*
B. Tech Mechatronics Engineering

Central Reserve Police Force Public School (C.R.P.F) - Delhi, India *Jun 2023*
Senior Secondary - CBSE

VOLUNTEERING & LEADERSHIP

Volunteer (Ambassador) - Kri8ivity League 2025, IIT Delhi

- Managed a national-level STEM competition, validating robotics builds against rules, coordinating teams of different states and volunteers, and overseeing on ground events operations and logistics.
- Demonstrated continuous learning by rapidly mastering new robotics protocols and event management software to improve competition logistics and team coordination efficiency.

CERTIFICATIONS & ACHIEVEMENTS

Data Science and Machine Learning - IIT Guwahati

A level examination in NCC Cadet Corps

Smart India Hackathon (SIH)-2024

HackWithMAIT 5.0

SKILLS

Python, C/C++, TypeScript, SQL, JavaScript, FastAPI, Flask, REST APIs, WebSockets, Authentication, JSON Schema Validation, PostgreSQL, SQLite, TimescaleDB, Pandas, NumPy, scikit-learn, LLMs, Agentic AI, RAG, LangChain, LangGraph, MCP, LLM APIs, ChromaDB, Ollama, Tool Calling, Git/GitHub, Docker, Docker Compose, Linux, Pytest, Automated Testing, CI/CD, Data Structures & Algorithms, Object-Oriented Programming

Languages: Python, C/C++, TypeScript, SQL, JavaScript
Backend & APIs: FastAPI, Flask, REST APIs, WebSockets, Authentication, JSON Schema Validation
Databases & Data/ML: PostgreSQL, SQLite, TimescaleDB, Pandas, NumPy, scikit-learn
AI/Agent Systems: LLMs, Agentic AI, RAG, LangChain, LangGraph, MCP, LLM APIs, ChromaDB, Ollama, Tool Calling
Engineering & Tools: Git/GitHub, Docker, Docker Compose, Linux, Pytest, Automated Testing, CI/CD
Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming

PROJECTS

Repo Pilot - AI Software Engineering Agent

- Built a 13-step AI software engineering pipeline incorporating automated root-cause analysis and problem resolution loops, demonstrating advanced problem-solving by autonomously diagnosing and repairing code errors.
- Engineered an event-driven FastAPI backend with local/Docker execution environments, WebSockets, and Git-based version control integration, including automated commit/push/PR workflows backed by 569 passing tests.

AI-Powered Customer Support Agent

- Built an AI-powered customer support agent combining intent classification, TF-IDF retrieval over 106K+ historical cases, grounded reply generation, and automated escalation for evidence-backed customer responses.
- Designed an evaluation pipeline with a 200-case human labelled benchmark, automated baseline comparisons, and blind human/LLM reply evaluation; achieved 57.5% auto-handle coverage while maintaining safety-focused escalation policies.